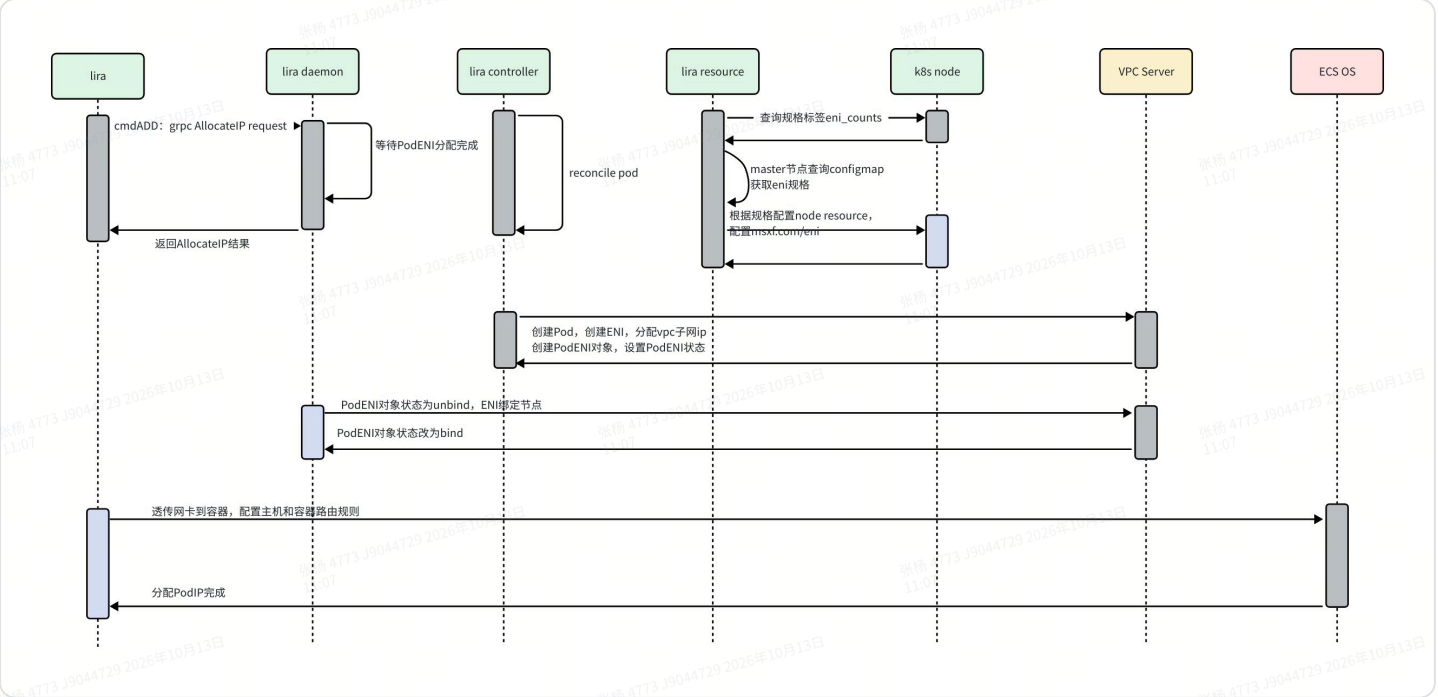
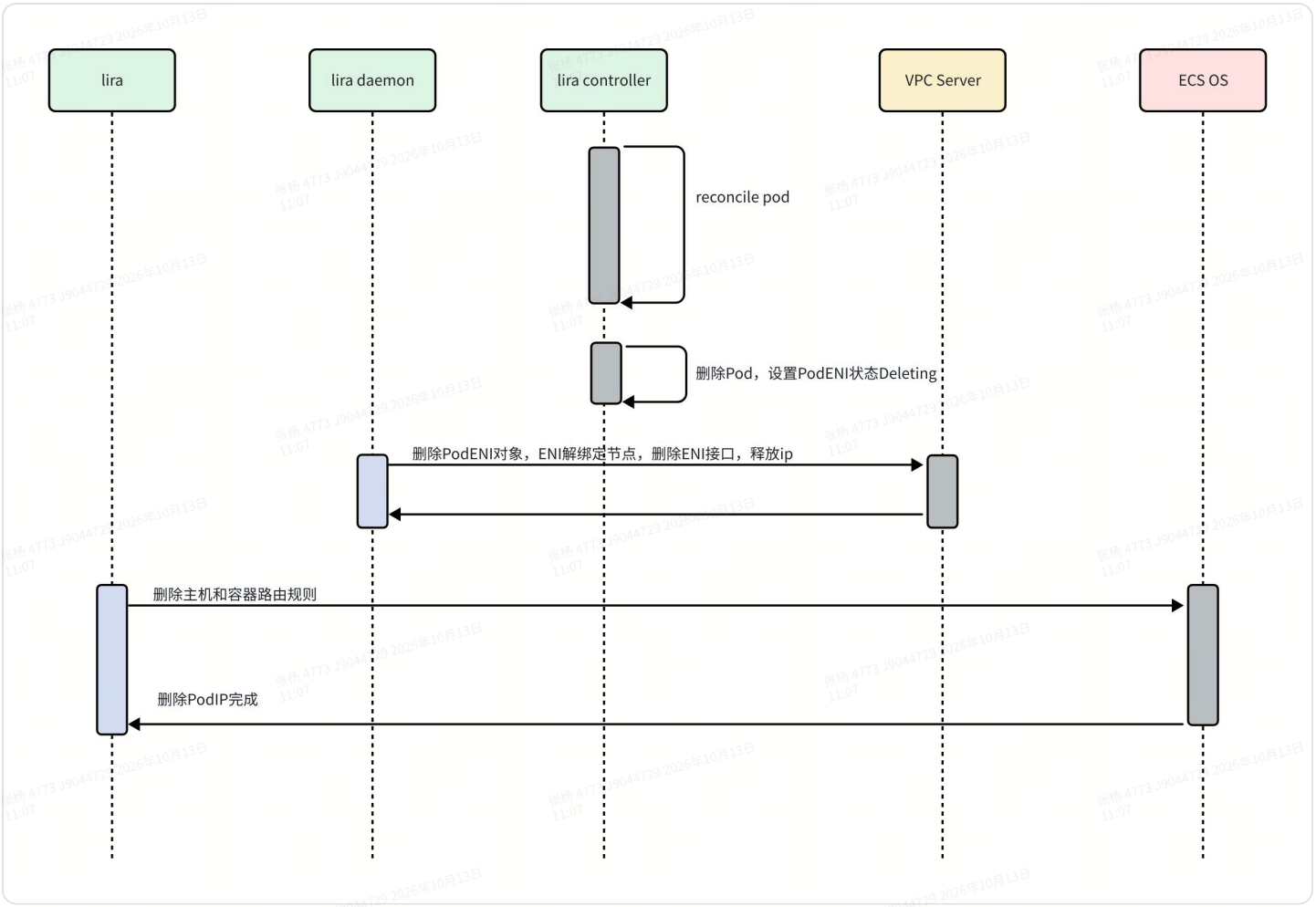
Lira cni插件问题定位手册

1. 流程图

1.1 分配ip流程图



1.2 删除ip流程图



2. 常用方式

cni分配ip失败，通常可以先看pod错误原因消息

代码块

```
kubectl describe po -n {ns} {podname}
```

通过返回消息判断是，daemon是否返回，还是podip已经分配，主机配置端口route信息报错。如果是daemon没返回说明podeni资源状态没正常，需要分析controller日志。如果daemon正常返回了，可能是配置节点网络设备报错需要分析容器运行时日志。mck下就是查看containerd日志。

查询podeni列表

代码块

```
kubectl get podeni -A
```

根据返回的podeni列表状态判断eni是否分配，podeni阶段分为：

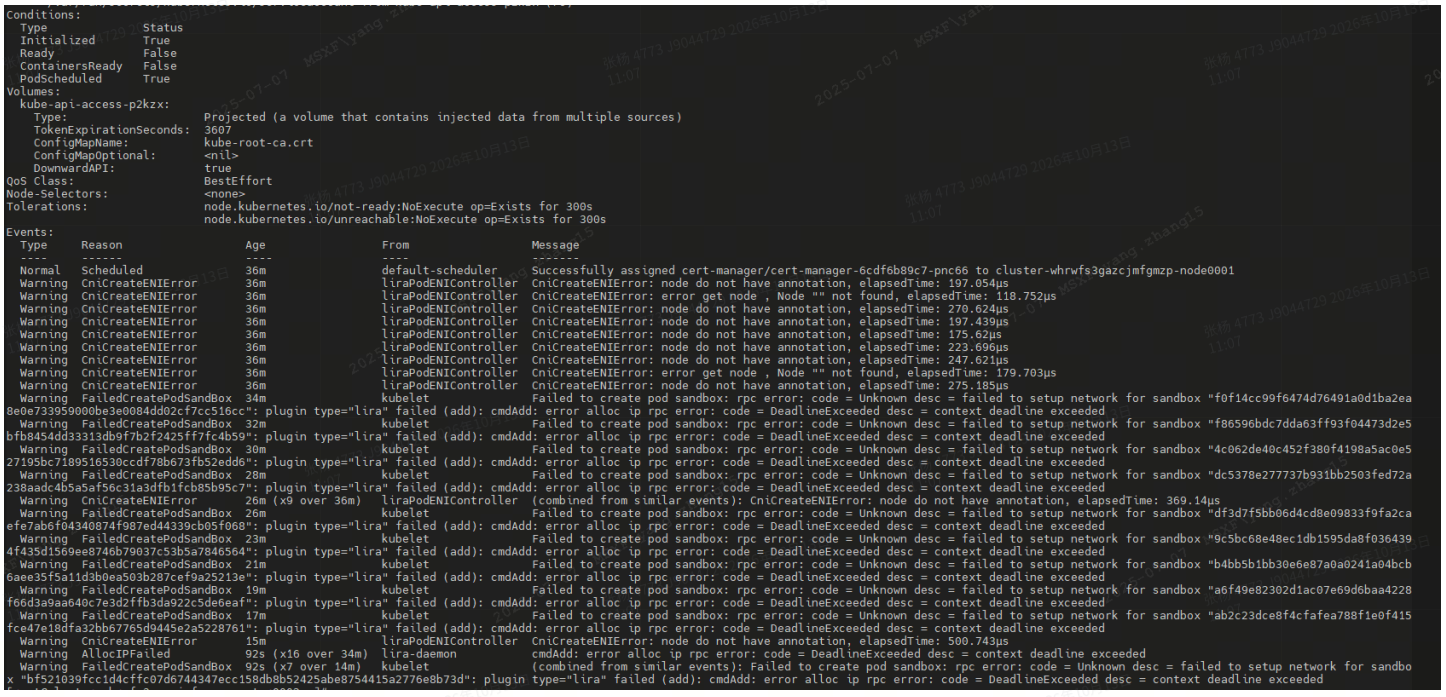
代码块

```
""       -- 空串表示初始状态
Created  -- eni已创建
Bind     -- eni已绑定
Deleting -- eni删除中
```

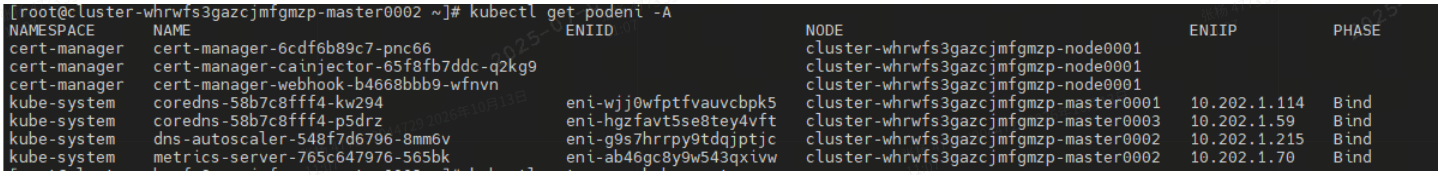
pod能使用的eni状态是Bind状态。

3. 问题分析案例

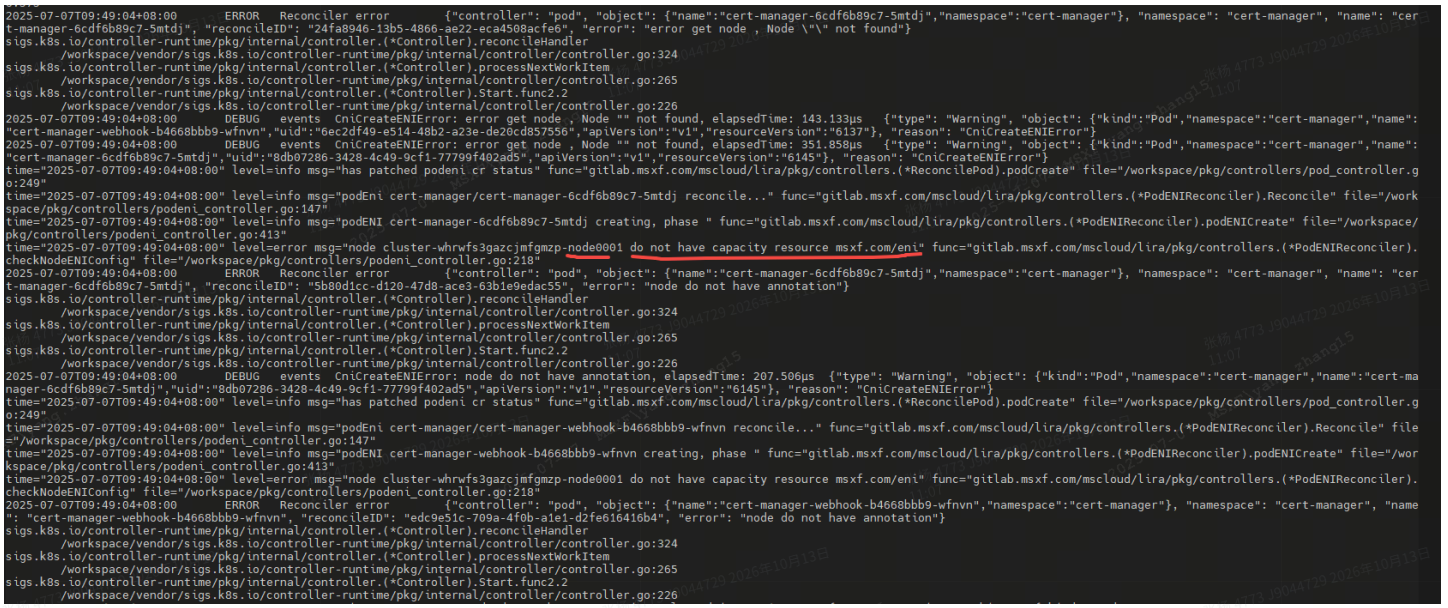
3.1 pod ip分配失败



可以看到lira cni分配ip cmdadd执行超时，就是发送grpc请求到lira daemon等待返回超时。lira daemon只会返回podeni状态为bind的对象。下面需要查看podeni列表：



可以看到node001上的3个podeni都没进入创建eni阶段。需要看下具体原因，podeni是由lira controller处理，查看日志lira controller：

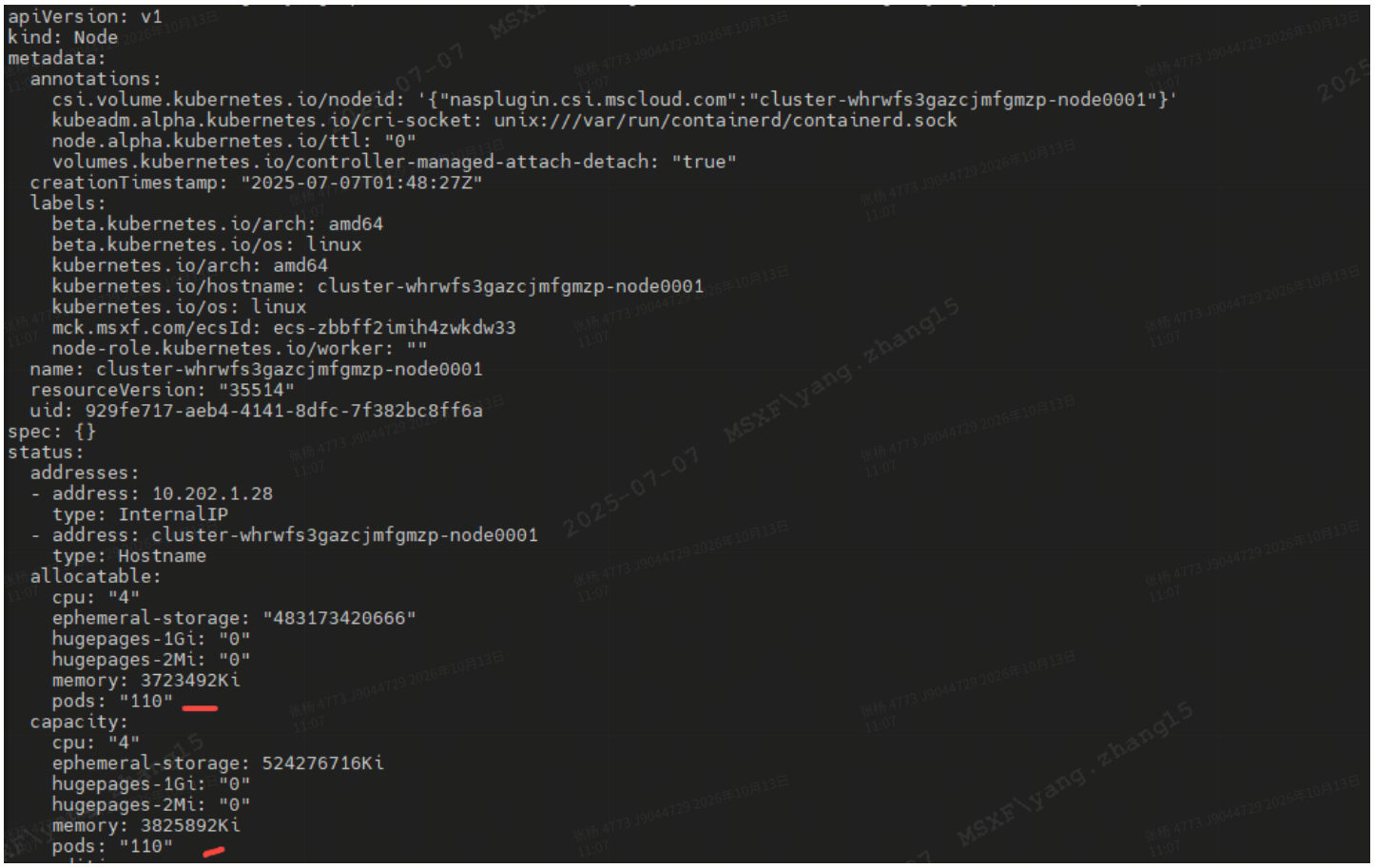


报错原因是cluster-whrwfs3gazcjmfgmzp-node0001节点上没有resource字段msxf.com/eni，资源字msxf.com/eni是lira resource注入到节点对象上的。节点上缺少msxf.com/eni一般是节点的label没打上。

```go
func updateNodeENIResource(ctx context.Context, msconfig *controllers.MSConfig) error {
    eniNums := "0"

    if utils.IsMasterNode(nodeCopy) {
        eniNums = msconfig.MasterENINum
    } else {
        //如果 types.ENI_COUNTS 存在于 nodeCopy.Labels 中，则 eniCountsErr 为 nil 或 false，表示操作成功
        eniCounts, eniCountsErr := nodeCopy.Labels[types.ENI_COUNTS]
        setupLog.Info("printeniCounts ", "printeniCounts", eniCounts)
        setupLog.Info("printeniCounts ", "eniCountsErr", eniCountsErr)
        if !eniCountsErr {
            msg := "node label eni_counts is missing or empty"
            labelMissingErr := errors.New(msg)
            setupLog.Error(labelMissingErr, "label is not exist")
            return labelMissingErr
        }
        eniNums = eniCounts

    }
    _, e := nodeCopy.Status.Capacity[corev1.ResourceName(types.ENIResourceKey)]      张杨15, 4 months ago • add files
    if !e {
        setupLog.Info("start patch node", "name", node.Name)
        // master 和 work节点区分，只变更work节点
        nodeCopy.Status.Allocatable[corev1.ResourceName(types.ENIResourceKey)] = resource.MustParse(eniNums)
        nodeCopy.Status.Capacity[corev1.ResourceName(types.ENIResourceKey)] = resource.MustParse(eniNums)
```



```
apiVersion: v1
kind: Node
metadata:
  annotations:
    csi.volume.kubernetes.io/nodeid: '{"nasplugin.csi.mscloud.com":"cluster-whrwfs3gazcjmfgmzp-node0001"}'
    kubeadm.alpha.kubernetes.io/cri-socket: unix:///var/run/containerd/containerd.sock
    node.alpha.kubernetes.io/ttl: "0"
    volumes.kubernetes.io/controller-managed-attach-detach: "true"
  creationTimestamp: "2025-07-07T01:48:27Z"
  labels:
    beta.kubernetes.io/arch: amd64
    beta.kubernetes.io/os: linux
    kubernetes.io/arch: amd64
    kubernetes.io/hostname: cluster-whrwfs3gazcjmfgmzp-node0001
    kubernetes.io/os: linux
    mck.msxf.com/ecsId: ecs-zbbff2imih4zwkdw33
    node-role.kubernetes.io/worker: ""
  name: cluster-whrwfs3gazcjmfgmzp-node0001
  resourceVersion: "35514"
  uid: 929fe717-aeb4-4141-8dfc-7f382bc8ff6a
spec: {}
status:
  addresses:
  - address: 10.202.1.28
    type: InternalIP
  - address: cluster-whrwfs3gazcjmfgmzp-node0001
    type: Hostname
  allocatable:
    cpu: "4"
    ephemeral-storage: "483173420666"
    hugepages-1Gi: "0"
    hugepages-2Mi: "0"
    memory: 3723492Ki
    pods: "110"
  capacity:
    cpu: "4"
    ephemeral-storage: 524276716Ki
    hugepages-1Gi: "0"
    hugepages-2Mi: "0"
    memory: 3825892Ki
    pods: "110"
```

工作节点缺少msxf.com/eni。

最终原因是节点上规格label没打上，mck-portal没把节点标签打上，导致eni创建失败，pod分配ip失败。